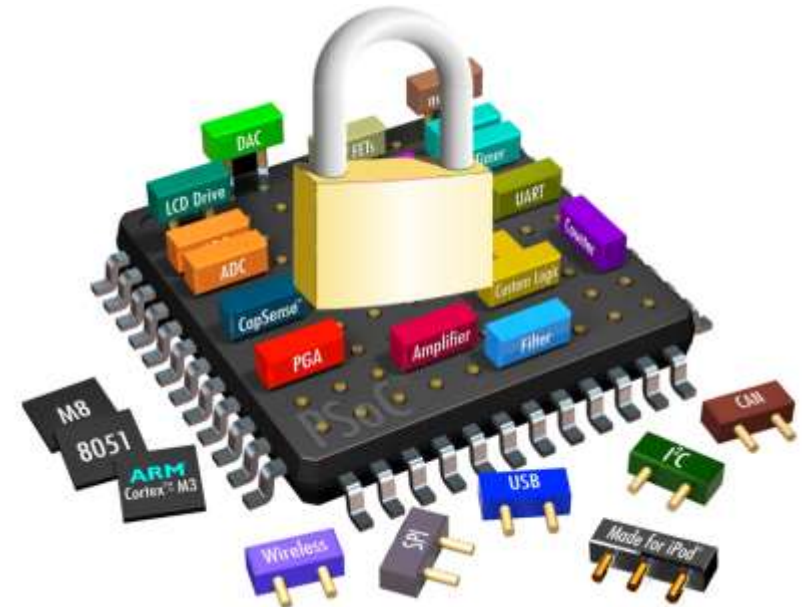


Hardware Security

TEE Examples: Keystone



Motivation

The Problem

Current trusted execution environments (TEEs) from major vendors—Intel SGX, ARM TrustZone, AMD SEV—represent **fixed** design points. Each makes inflexible choices in deployability, trusted computing base (TCB), and threat models.

Platform providers and developers are locked into limited possibilities, unable to customize for their specific needs. SGX's 128 MB memory limit causes drastic slowdowns in large-scale applications. Proprietary ISAs make modifications nearly impossible.

The Vision

We need **customizable TEEs**—a paradigm shift similar to Software Defined Networking. Hardware manufacturers provide baseline standards, platform providers customize implementations, and developers choose features that match their requirements.

This enables quick prototyping, faster fixes, adaptation to new threats, and use-case specific deployment without rebuilding from scratch.

Core Design Principles

01

Hardware-Enforced Memory Isolation

Flexible isolation as a native primitive, customizable at runtime by privileged mode

03

Modularity Throughout

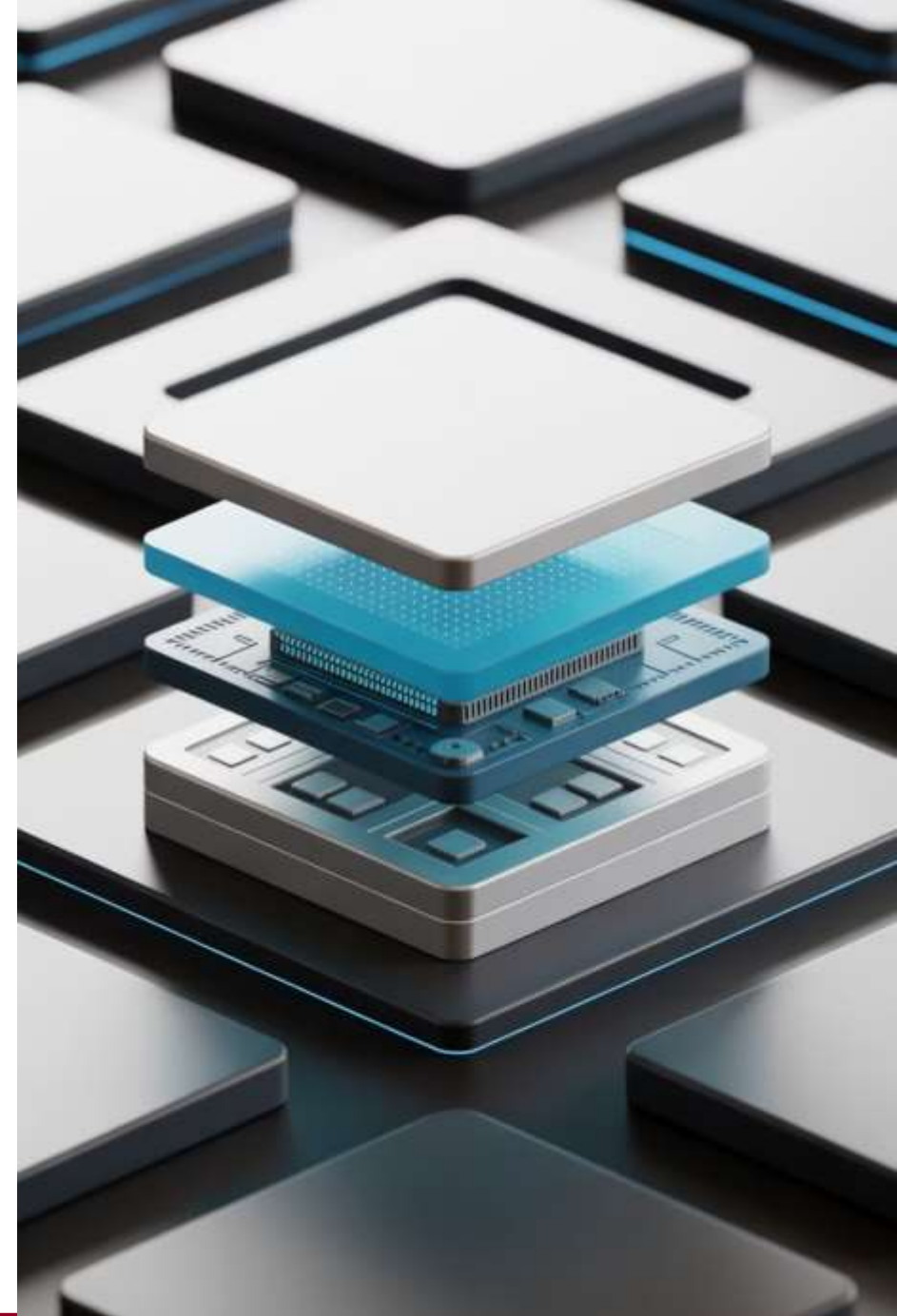
Independent layers with clean abstractions, maintaining compatibility with existing privilege levels

RISC-V provides these abstractions through M-mode (programmable highest privilege), hardware interrupt delegation, and PMP (runtime memory isolation). This makes RISC-V a natural fit for customizable TEEs.

02

Programmable Trusted Layer

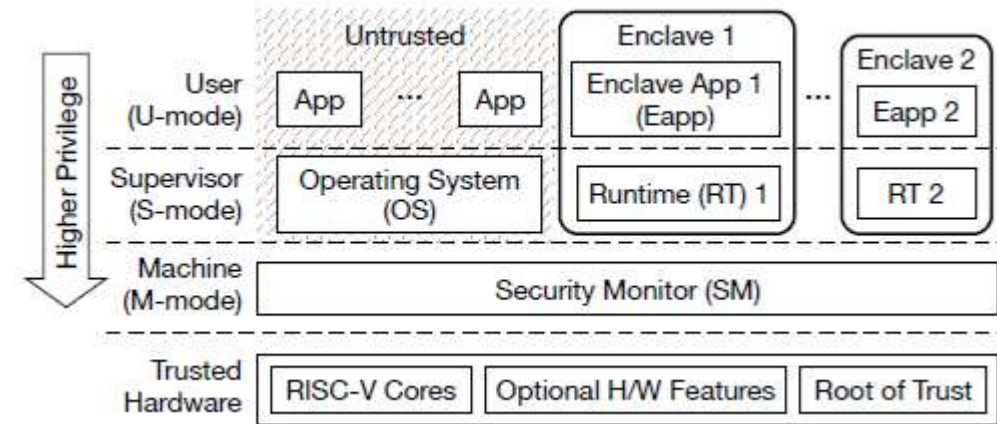
Software layer with higher privileges than OS, managing isolated memory regions with minimal TCB



Customization in Action

- Platform Support
 - Unlike vendor-specific TEEs, Keystone supports various workloads across embedded devices, mobile platforms, and cloud servers without platform lock-in
- TCB Size
 - Modular design allows minimal TCB for simple applications or rich functionality for complex ones—same database can run with thousands or millions of lines of code
- Threat Models
 - Flexible defense mechanisms adapt to physical attackers, software attackers, side-channel attacks, or denial-of-service—enable only what you need

System Architecture

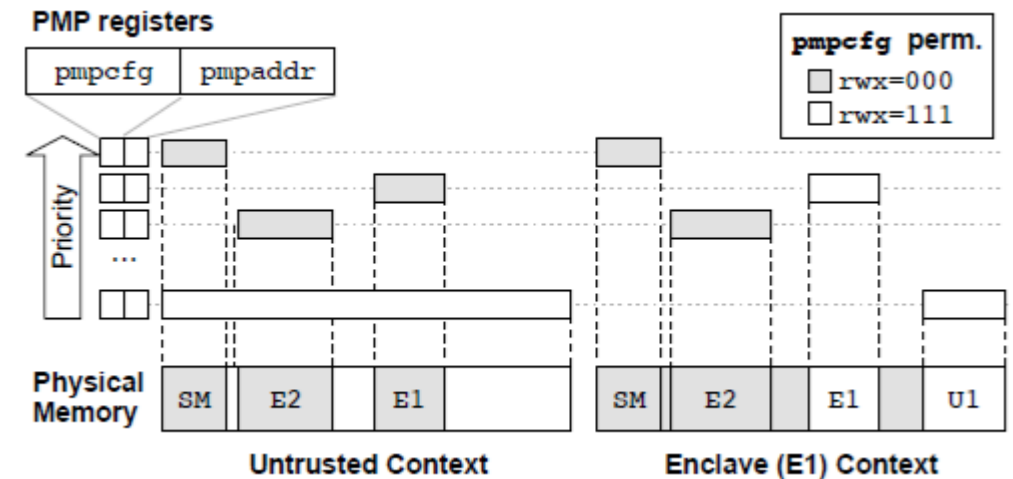


- Security Monitor (SM)
 - Executes in M-mode, enforces TEE guarantees across the platform.
 - Configurable by the platform generator.
- Runtime (RT)
 - Executes in S-mode within enclave, manages lifecycle and memory.
 - User-specified and configurable per application needs.
- Enclave Application
 - Executes in U-mode, isolated from untrusted OS.
 - Can be unmodified existing applications with appropriate RT.

Memory Isolation with RISC-V PMP

Keystone leverages RISC-V Physical Memory Protection (PMP) for flexible, dynamic memory isolation. PMP allows M-mode to restrict S-mode and U-mode access to customizable physical memory regions.

- 1 Flexible Regions**
Multiple contiguous enclave regions proportional to PMP registers—no single large shared region
- 2 Arbitrary Sizes**
Regions from 4 bytes to entire DRAM with page-aligned memory of any size
- 3 Dynamic Reconfiguration**
Create or release regions during execution instead of permanent occupation



The SM configures PMP entries to protect its own memory and enclave regions. During context switches, permission bits flip to allow enclave access while protecting OS memory, and vice versa.

TEE Primitives

- Secure Boot
 - Root-of-trust measures SM image, generates attestation key, signs measurement with hardware secret
- Secure Randomness
 - Hardware source or fallback methods (CPU jitter, DRAM decay) via SM SBI interface
- Remote Attestation
 - SM signs enclave measurement with attestation key, binds to secure channel construction
- Sealed Storage
 - Monotonic counters in SM memory, future support for encrypted content tied to enclave identity
- Trusted Timer
 - Non-writable hardware timer registers accessible via rdcycle instruction
- Secure Delegation
 - Safe trap handling without leaking enclave state or allowing arbitrary entry points